Position, Attention Geometry, and Controlled Evidence Routing for Retrieved Native-KV Memory

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Abstract

Chunked key/value (K/V) memory introduces a positional ambiguity that ordinary prefix caching largely avoids. A token can belong at position 100 in its source yet be encoded at position zero because its chunk was processed independently. The resulting tensor is valid, but it no longer represents the source coordinate or attention relation that retrieval later assumes. We study this failure in Progressive Retrieval Attention (PRA), which uses compact gists to select chunks and then attends to their full-detail native K/V.

A source-relative logical offset is the general repair. Across learned absolute, sinusoidal, and rotary position embedding (RoPE) models, offsets remove layer-0 reset error exactly and reduce final-layer K error in all 30 paired comparisons over two capacities and five seeds. Overlap or grouped encoding addresses the separate loss of causal context, at additional encoding cost. RoPE adds a narrower capability: pre- and post-RoPE keys are exactly equivalent at matched effective positions, while pre-RoPE storage permits retrieval-time rebinding.

The central result is a separation rather than a universal quality gain. WikiText-2 representation improves in 30/30 pairs, but next-token loss improves in only 5/30; controlled QA-derived loss improves in 35/60 pairs and reverses direction for some RoPE conditions. Changing RoPE distance alone yields no stable useful optimum. Increasing centered-RoPE gist resolution from one to eight raises Spearman agreement with our defined native-Q/K chunk-ranking surrogate from .794 to .963 while normalized target-identity recall/request-fraction area remains .6510–.6516. Conversely, a learned 32-dimensional metric reaches .971 area with only .178 surrogate rank fidelity on explicit answer-code anchors that always occupy the first of 15 candidates. This controlled dissociation does not establish general semantic relevance or downstream answer improvement. For fixed selected identities, post-RoPE K, V, and native attention output remain bit-exact. A 192-token logical prompt also runs with every measured native operation at or below 24 tokens under a 32-token limit. The distinction remains consequential in a pretrained Qwen3 8B–32B MLX calibration: at 8K, source-relative segmented first-logit error is far below a query-restart control, whose greedy-sequence agreement is zero at every measured model point. Positional continuity is necessary, but coordinate correctness, representation fidelity, attention geometry, controlled target selection, and downstream use are distinct empirical properties.

1 Introduction

A key retrieved from external memory carries more than content. Its hidden state reflects the causal context present when it was encoded, and its attention score depends on a position. If a long source is divided into bounded blocks and every block starts again at zero, later tokens silently inherit the coordinates of earlier ones. The cache remains a well-formed tensor, but it misrepresents the source.

This problem is broader than RoPE. Learned absolute embeddings and sinusoidal embeddings also change when a block is reset. RoPE is special for a different reason: because position is an explicit rotation of queries and keys, it provides an algebra for preserving or changing a cached key's binding. The offset requirement is general; the rebinding operation is rotary.

We study both problems in PRA, a sparse native-KV memory mechanism. A long source or displaced prompt head is encoded in bounded blocks. Compact gists route a later query to selected chunks, but attention consumes the chunks' original token-level K/V. The position carried by those keys must therefore be a cache contract rather than an implementation accident.

We organize the problem into five increasingly demanding levels. Retrieval selection and downstream use are separate because a router can find annotated evidence while the model still fails to use that composition:

1. **Coordinate correctness:** were source-relative positions preserved?
2. **Representation fidelity:** does chunked K/V resemble the state produced with the intended coordinates and context?
3. **Attention-geometry fidelity:** does the positional treatment reproduce the defined query–key chunk-ranking relation?
4. **Target-identity retrieval utility:** is the programmatically labeled chunk ranked highly by a routing representation that applies no explicit positional rotation at scoring time?
5. **Downstream task utility:** does the selected state improve language-model loss or answers?

These levels form a dependency, not an equivalence. Correct coordinates do not recreate missing causal context. Faithful K/V can still be selected poorly. Even annotated evidence can be an inferior context composition for a particular model. The pooling and routing experiments make one boundary especially clear:

> **The geometry appropriate for executing native attention is not necessarily the geometry appropriate for searching memory.**

The paper makes eight outcome-oriented contributions:

- We formalize source-relative logical position as cache provenance for independently encoded K/V.
- We show that logical offsets repair reset fragmentation for learned absolute, sinusoidal, and RoPE models, rather than only for rotary models.
- We verify exact pre-/post-RoPE equivalence and quantify the runtime cost of deferred rotation and retrieval-time rebinding.
- We separate coordinate repair from overlap-based contextualization and measure the quality–encoding-cost tradeoff.
- We demonstrate that systematic representation improvement need not produce monotonic language-model or QA-derived loss improvement.
- We show that averaging post-RoPE keys mixes positional phases, and that increasingly faithful approximations of a defined native-Q/K chunk-ranking surrogate need not improve programmatic target-identity retrieval.

- We demonstrate, on a deliberately simplified fixed-anchor task, that a tiny learned hidden-state projection can improve target-identity ranking while selected native post-RoPE K/V remains unchanged for attention. We do not interpret this as general semantic retrieval.
- We demonstrate bounded implicit-head execution in which logical prompt length exceeds every native model operation, while carefully separating this mechanical result from dense long-context equivalence.

2 What Position Does a Retrieved Token Own?

Suppose a source span occupies positions 100 through 131. Independent chunk encoding often does this:

View	First token	Last token
Source/physical position	100	131
Wrong chunk-local reset	0	31
Correct logical/effective position	$100 + 0$	$100 + 31$

The correction is simply

$$p_{\text{effective}} = p_{\text{chunk,start}} + p_{\text{inside}}, \tag{1}$$

where $p_{\text{chunk,start}}$ is stored once for a contiguous chunk and $p_{\text{inside}} \in [0, T-1]$ is its local index. The formula restores the coordinate the token owns inside its source.

The word *source-relative* is important. Positions 100–131 are meaningful within one document, reference, or displaced prompt head. Two unrelated documents do not acquire a meaningful global ordering merely because both are registered in one request. Source identity and logical offset are therefore cache provenance. A continuously advancing `#__head`, by contrast, represents one prompt history and retains one coordinate system as direct tokens roll into memory.

3 What Does PRA Store and Retrieve?

PRA is self-contained in the following path:

$$\boxed{\text{reference or old prompt}} \rightarrow \boxed{\text{bounded chunks}} \rightarrow \boxed{\text{routing gists}} \rightarrow \boxed{\text{selected chunks}} \rightarrow \boxed{\text{full-detail K/V}} \rightarrow \boxed{\text{bounded attention}}.$$

The gist is routing metadata, not a compressed substitute for the selected token state. Once a chunk is selected, its full token-level native K/V joins direct-context K/V under the same attention softmax. Position errors introduced while building that K/V survive routing and materialization; the router cannot repair them afterward. For one query head and a router-selected identity set S, PRA executes the ordinary attention map

$$\mathrm{Attn}(q, K_S, V_S) = \mathrm{softmax}\left(\frac{qK_S^\top}{\sqrt{d_h}} + M_S\right) V_S, \tag{2}$$

where K_S, V_S are the selected tokens' native payload, d_h is head width, and M_S enforces the applicable causal mask. The compact gist determines S but is not substituted for K_S or V_S.

Three lengths remain distinct. L_{logical} is the addressable source span, $L_{\text{model-op}}$ is the maximum K/V participating in one native operation, and L_{position} is the supported coordinate range. PRA can make $L_{\text{logical}} > L_{\text{model-op}}$ by caching and selecting bounded regions, but it does not by itself guarantee that a model has learned useful behavior at every L_{position}.

Table 1 gives the empirical answers before the details. Each row addresses a different question; no row certifies the one below it.

Table 1: Headline results. Each row answers a different correctness question.

Question	Verified result	Meaning
Do offsets repair reset coordinates?	Layer-0 K RMSE becomes zero in 30/30 paired comparisons	Source-relative continuity is a general requirement
Do deeper representations improve?	Final-layer K RMSE falls in 30/30 pairs	The repair survives two capacities and three mechanisms
Does overlap improve representation?	Yes in the verified WikiText scope; 29/30 loss pairs also improve modestly	Contextualization is separate and costs 1.94× encoding at 8L
Are pre/post RoPE keys equivalent?	Zero K, logit, and output error at matched effective positions	Storage state does not change the intended geometry
Does better representation guarantee task quality?	No: WikiText loss improves in 5/30; QA routed loss in 35/60	Functional utility is a separate empirical property
Can deferred rotation rebind keys?	Yes; warm latency rises from 0.179 to 0.649 ms	Flexibility has a measured systems cost
Does closer native-Q/K surrogate geometry improve target retrieval?	Centered $G = 1 \rightarrow 8$ raises surrogate rank fidelity .794→.963 while target-identity area stays .6510→.6516; the learned 32-D metric reaches .971 area at only .178 fidelity	The tested chunk-ranking and target-identity objectives dissociate
Can logical prompt exceed a native operation?	192 logical tokens, maximum measured operation 24 under a limit of 32	Bounded positional and transport mechanics

The mechanisms can now be stated without conflating their roles:

Table 2: Five levels of memory correctness and utility. Controlled evidence at each boundary shows why success at one level does not certify the next.

Level	Controlled evidence	Boundary exposed
Coordinate correctness	Offsets remove layer-0 reset error in 30/30 pairs.	Correct coordinates do not reconstruct missing causal context.
Representation fidelity	Final-layer K improves in 30/30 pairs; WikiText loss improves in 5/30.	Representation fidelity is not task utility.
Attention-geometry fidelity	Centered $G = 1 \rightarrow 8$ raises surrogate rank fidelity .794→.963 while target-identity area stays .6510→.6516.	Agreement with this chunk-ranking surrogate is not target-identity retrieval.
Target-identity retrieval utility	The learned metric reaches .971 area with .178 surrogate rank fidelity; an evidence oracle can still worsen loss.	Programmatic identity recall is not sufficient for downstream use.
Downstream task utility	Routed QA loss improves in 35/60 pairs, with reversals across RoPE cells.	Context composition and model use require direct evaluation.

4 Do Logical Offsets Repair More Than RoPE?

The general experiment compares independently encoded 32-token blocks with a dense 128-token representation. It covers two model capacities, learned absolute, sinusoidal, and RoPE positions, and five matched seeds per condition. Table 3 reports layer-0 and final-layer K RMSE.

Table 3: Reset versus source-relative offsets. Every final-layer paired difference favors offsets; every offset layer-0 error is exactly zero.

Tier	Position	Reset L0	Offset L0	Reset final	Offset final	Improved
Tiny	Learned absolute	.6673	.0000	.6769	.1129	5/5
Tiny	Sinusoidal	.3640	.0000	.4722	.1436	5/5
Tiny	RoPE	.6463	.0000	.6745	.1373	5/5
Small	Learned absolute	.6626	.0000	.7193	.3066	5/5
Small	Sinusoidal	.3345	.0000	.5507	.2746	5/5
Small	RoPE	.5586	.0000	.7498	.4235	5/5

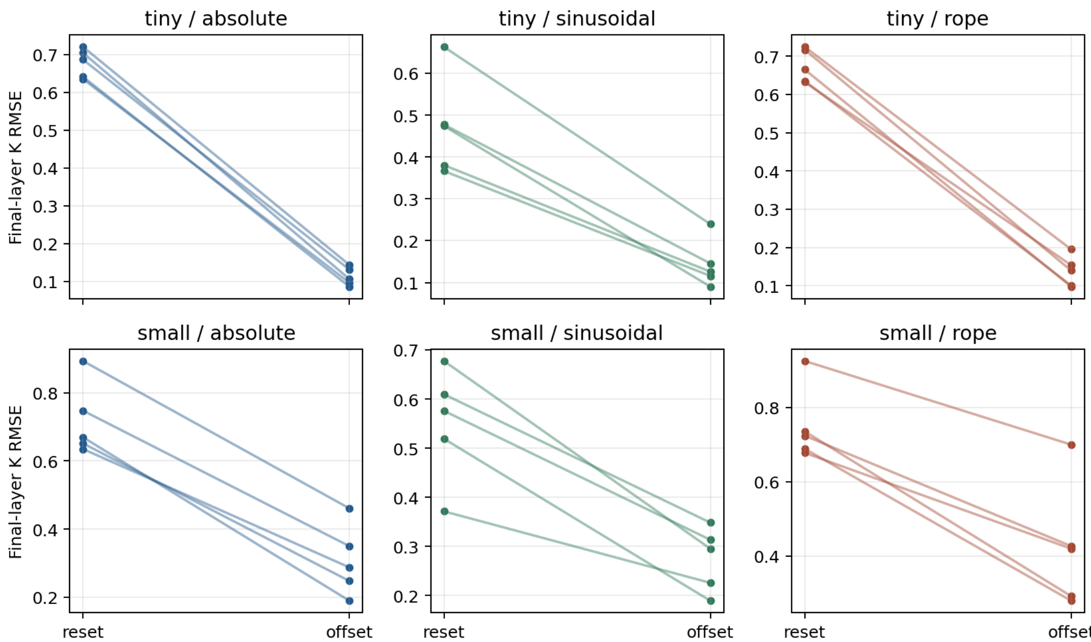

Figure 1: Offsets reduce final-layer K error in all 30 paired runs. The residual after exact layer-0 repair reveals a second problem: missing causal context across encoding blocks.

The sinusoidal result rules out a RoPE-only explanation. Logical offsets repair coordinate reset because they restore source position, regardless of whether position enters through an embedding table, a formula-defined additive embedding, or rotary Q/K geometry.

The residual final-layer error is equally informative. At layer zero, matched token content and effective positions suffice for exact equality. At later layers, a block lacks the causal left context available in dense encoding. This is contextual fragmentation, not failed coordinate repair.

5 What Does Overlap Repair?

Encoding overlap gives each block additional native left context and commits only its new core. It does not change routing granularity and should not be described as a routing intervention. In the controlled representation test, 25% overlap lowers final-layer mean error in all six tier–mechanism conditions. At 50%, five of six improve; small learned absolute RMSE moves slightly from 0.3066 to 0.3103. The intervention is useful but not monotonic.

WikiText-2 gives a clear quality/cost point. At an 8:1 logical-to-native ratio, overlap lowers final-layer error in every condition and next-token loss in 29/30 paired runs relative to offset-only encoding. The loss reductions are modest, from 0.0007 to 0.0173 in the tiny tier and 0.0102 to 0.0148 in the small tier, while encoding processes 1.9375 times the unique source tokens. Overlap buys contextualization through recomputation.

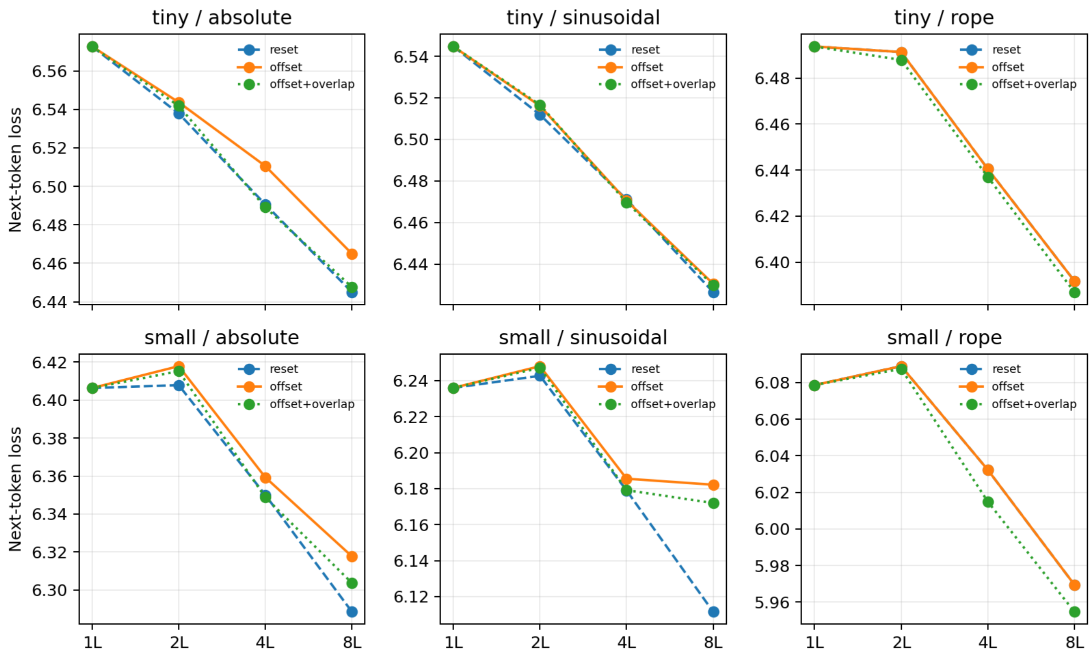

Figure 2: On WikiText-2, offsets sharply improve representation fidelity without a matching loss gain. Overlap gives smaller, more consistent loss changes at almost 1.94× encoding cost at 8L.

Across the QA-derived probes, overlap effects are mixed and mechanism-dependent. That does not make overlap a failed coordinate method; it confirms that it repairs a different input to the task pipeline.

6 What Does RoPE Uniquely Add?

6.1 Intuition Before Algebra

RoPE rotates each query and key according to its position. Attention sees the difference between those rotations, so translating both a query and key together preserves their relative geometry. Translating only the key changes its distance from a fixed query and is not an invariant operation.

This yields two storage choices:

Post-RoPE K	Pre-RoPE K
Position is bound when the cache is written. Retrieval is cheap, but the binding is fixed.	Position is applied when the cache is read. Retrieval-time rebinding is possible, with extra work.

6.2 The Minimal Algebra

For a head of width d_h, let R_p denote the RoPE rotation at position p. The rotated query and key are

$$\widetilde{q}_i = R_i q_i, \qquad \widetilde{k}_j = R_j k_j. \tag{3}$$

Because $R_i^\top R_j = R_{j-i}$,

$$\widetilde{q}_i^\top \widetilde{k}_j = q_i^\top R_{j-i} k_j. \tag{4}$$

Equation 4 answers one question: why does common translation preserve the query–key relation? Adding c to both positions leaves $j - i$ unchanged. It does not say that $R_i q_i$ has the same relation to $R_{j+c} k_j$.

Pre- and post-RoPE storage are algebraically equivalent when they use the same effective position:

$$K_{\text{post}}(p) = R_p K_{\text{pre}}. \tag{5}$$

The four-cell model test confirms zero K, logit, and output RMSE, with top-token agreement 1.0, for matched offset positions across both tiers, all layers, and five seeds. Reset binding instead has mean K RMSE 0.648 and output RMSE 0.063. Deliberate rebinding by +512 changes the output, as it should.

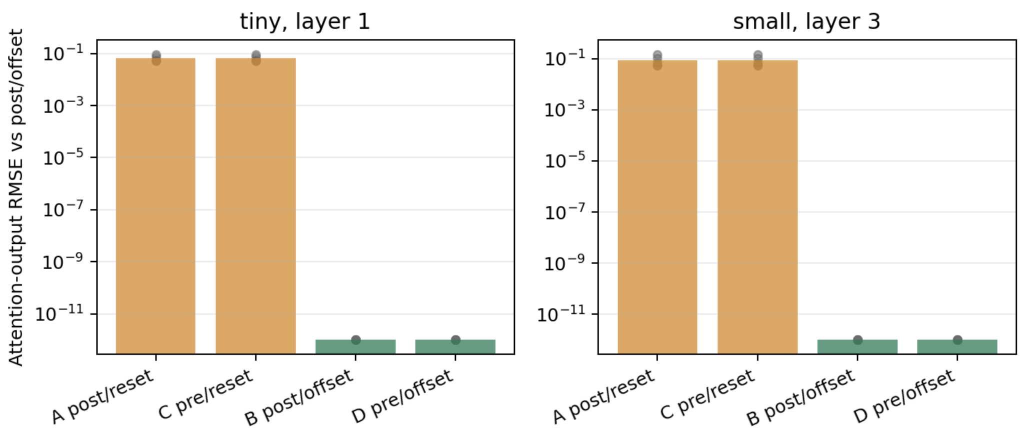

Figure 3: Matched pre- and post-RoPE storage reproduces the same attention geometry exactly. Reset coordinates and deliberate K-only rebinding change it.

The pure tensor test reaches the same conclusion. Translating Q and K together preserves the top-attended token, with maximum logit RMSE 5.79×10^{-5} through the measured shifts. Moving K alone by up to 8,192 positions changes fixed-query scores. This is the correct behavior, not an error in RoPE.

6.3 Deferred Rotation Has a Cost

At 160 selected small-tier tokens, storing post-RoPE K gives a 0.1791 ms warm path. Storing pre-RoPE K, reconstructing positions, and rotating at materialization gives 0.6487 ms. The position reconstruction itself is small; the unfused RoPE transform dominates. The nearly flat timing across token counts indicates launch-dominated behavior on the measured GTX 950M, so this is a prototype cost measurement rather than a production throughput prediction.

Both modes retain the same number of K/V elements. For 192 stored small-tier tokens, K/V uses 196,608 bytes. A contiguous chunk needs one 8-byte logical start for execution; a full 192-token int64 audit vector would use 1,536 bytes. Deferred rotation changes when phase is assigned, not the size of the native K/V payload.

6.4 Retrieval-Time Placement Preserves Local Geometry

Deferred rotation exposes a narrower control than an arbitrary position reset. Let a selected chunk have source positions $s_0, \ldots, s_{n-1}$ and let the current query be at q. We define the nearest-token distance D and translate the chunk as a unit:

$$s'_j = s_j + (q - D - s_{n-1})\,. \tag{6}$$

Then $q - s'_{n-1} = D$, while $s'_j - s'_\ell = s_j - s_\ell$ for every pair of tokens. The experiment therefore changes the global query–memory displacement but preserves token order and all within-chunk spacing. Exact placement leaves s_j unchanged; local placement sets $D = 1$; clipped placement uses $\min(D_{\text{source}}, D_{\max})$.

This distinction matters because RoPE does not simply lose a fixed amount of resolution with distance. It represents relative displacement through many periodic frequency pairs: high-frequency pairs vary rapidly and cycle often, while lower-frequency pairs vary more slowly. A displacement can be mathematically valid yet fall outside the distance distribution for which the learned Q/K projections became useful. Equation 6 tests that functional question without changing memory content.

7 Why Does Better Geometry Not Guarantee Better Loss?

The central empirical discovery appears when representation and task metrics are placed on the same axis. On 8L WikiText-2 windows, offsets reduce final-layer K RMSE in 30/30 paired runs but reduce next-token loss in only 5/30. Learned-absolute loss rises by 0.0200 in the tiny tier and 0.0292 in the small tier; sinusoidal loss rises by 0.0040 and 0.0707; RoPE loss is unchanged to the reported precision.

The controlled HotpotQA- and QASPER-derived answer-code probes likewise resist a universal story. Routed loss improves in 35/60 paired comparisons. Learned-absolute and sinusoidal means improve in both capacities and datasets, whereas RoPE means worsen in all four dataset–capacity cells. These probes use natural source text but simplified targets; they measure positional and retrieved-memory behavior, not unrestricted QA.

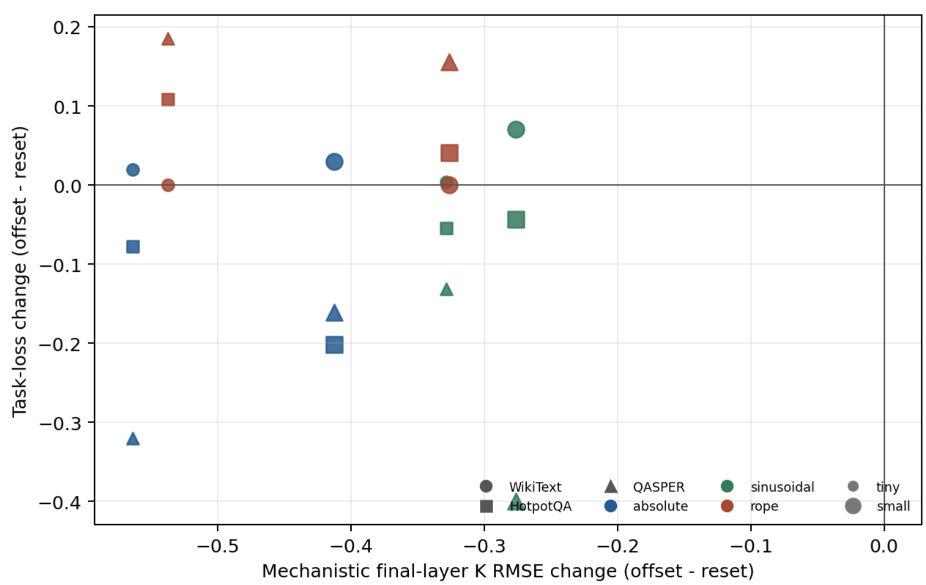

Figure 4: Every point moves left because offsets improve representation fidelity, but task-loss changes cross zero. Correct geometry is not a sufficient predictor of usefulness.

This is not a failed universal-gain result. A universal improvement would have hidden the fact that these are different properties. A model trained on short reset blocks may not have learned to use the repaired distance distribution. Routing gists may rank different chunks after positions change. Selected evidence may lack surrounding support, and distractors can alter the model's computation. Position repair makes the state better defined; it does not choose the state or teach the model how to use it.

8 Does RoPE Distance Explain Task Quality?

This section stays within level 3, attention-geometry fidelity. It asks a specific hypothesis: perhaps exact source-relative RoPE is geometrically correct but functionally suboptimal because retrieved memory is too distant from the query. Can virtual rebinding help? The task results also leave a contextual alternative. Very small memory fragments might lack enough contextual state, or exact source placement might expose a learned relative-distance mismatch. We separate encoding context from materialized-unit size and begin with oracle evidence so routing does not obscure the comparison.

8.1 Fixed Distance Changes Attention, but Has No Stable Optimum

The fixed-D sweep reuses the same raw K, V, query, evidence, and local chunk geometry for every policy. It evaluates $D \in \{32, 64, \ldots, 4096\}$ on tiny RoPE checkpoints over HotpotQA- and QASPER-derived probes, using five matched seeds. Exact nearest-token source distances average 87.3 and 90.4, respectively.

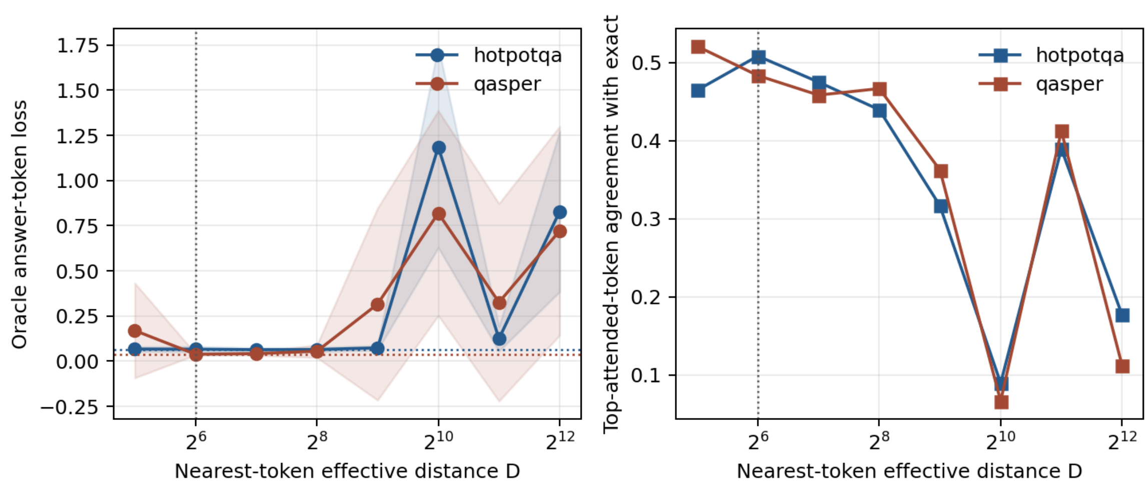

Figure 5: The same retrieved evidence under different RoPE displacements. Dotted horizontal lines show exact-placement loss; the vertical line marks the pooled selected value $D = 64$. Bands are one standard deviation over five seeds. Remote placement can sharply damage loss and top-token agreement, but the useful near range has no stable winner.

The sweep establishes that placement is not inert. At $D = 1024$, HotpotQA-derived mean loss rises from .0655 under exact placement to 1.1855, and QASPER-derived loss rises from .0377 to .8192. Agreement with the exact policy's top-attended token falls to .090 and .067. The response is not monotonic: $D = 2048$ is less harmful than $D = 1024$, consistent with RoPE's multi-frequency periodic geometry rather than a scalar distance-decay story.

The near-range result resolves the hypothesis negatively. In the two-tier confirmation matrix, fixed $D = 64$ beats exact placement in 50 of 100 matched dataset–tier–seed–composition cells and loses in 50; mean loss changes by $+8.1 \times 10^{-5}$. Clipping only source distances above 64 wins 68/100 cells, but its mean change is merely -9.3×10^{-5}. Log compression wins 24/100 and changes mean loss by $+1.3 \times 10^{-4}$. These effects are far smaller than the remote-distance failures and do not establish a task-optimal virtual distance.

8.2 Context and Materialized Composition Matter More

The encoding-context ablation jointly encodes one, two, or four adjacent references while retaining five routing units. Its evidence-only oracle row is invariant: the evidence span is causally first, so later references cannot alter its hidden state. This is a useful design check, not evidence that context never matters. When all encoded neighboring K/V is retained, small-tier HotpotQA-derived loss falls from .0235 with one reference per block to .0026 with four; QASPER-derived loss falls from .0936 to .0026. Tiny QASPER improves modestly from .0388 to .0380, while tiny HotpotQA is essentially flat.

The materialized-unit ablation keeps the encoding regime fixed and changes source partitioning. Exact-position oracle loss generally worsens as the source is divided more finely. HotpotQA-derived tiny loss is .0607, .0655, and .0775 for 2, 5, and 16 units; small-tier loss is .0026, .0030, and .0040. QASPER-derived small-tier loss similarly moves from .0026 to .0027 to .0036; tiny QASPER has a shallow five-unit plateau before rising to .0426 at 16 units. A larger selected unit includes more neighboring content, so this probe identifies a composition benefit rather than a pure representation-size law.

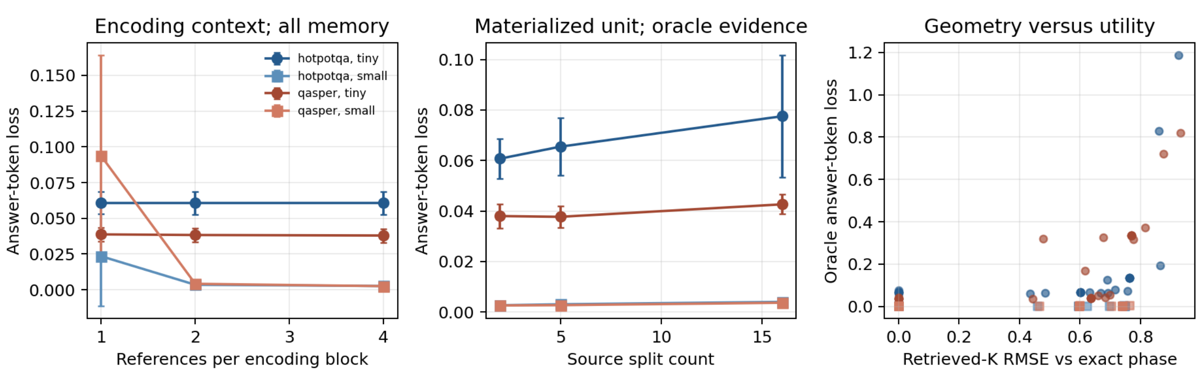

Figure 6: Left: larger encoding groups help when the resulting neighboring K/V is retained. Center: finer source partitioning generally raises oracle loss. Right: larger deviation from exact RoPE phase does not map monotonically to task loss. Error bars show one standard deviation over five seeds.

The combined diagnosis is therefore an interaction dominated here by contextualization and composition. Their effects are larger and more stable than the small near-range distance changes. Exact source placement is substantially better than a full local reset in the tiny-tier oracle rows, and a small near-query rebase does not reliably improve on exact placement. Remote rebinding can be destructive, but the measured models do not support a universal replacement geometry. For continuous history, exact source coordinates remain the default. For independent references, where no natural serialization distance exists, an oracle gate could compare exact and a conservative clipped policy as a diagnostic before general retrieval-time rebinding is enabled. The broad adaptive-geometry program is not justified by this fixed-D evidence.

9 Does Pooling Preserve Native RoPE Geometry?

Token-level post-RoPE keys are the correct native attention representation. Compact routing, however, usually replaces several token keys with one arithmetic mean. Native post-RoPE K is correct token by token, but arithmetic pooling combines vectors carrying different rotary phases. For pre-RoPE token keys K_j^{pre} at source positions p_j, native keys and their post-RoPE mean are

$$K_j^{\text{post}} = R(p_j)K_j^{\text{pre}}, \qquad g_{\text{post}} = \frac{1}{m}\sum_j R(p_j)K_j^{\text{pre}}. \tag{7}$$

A mean formed after rotation and a mean formed before rotation then bound to one representative center p_c are, in general,

$$\boxed{\frac{1}{m}\sum_j R(p_j)K_j^{\text{pre}} \neq R(p_c)\left(\frac{1}{m}\sum_j K_j^{\text{pre}}\right).} \tag{8}$$

The left side is a valid mean of individually correct native keys, but it has no single RoPE phase. The right side has one coherent representative phase, but compresses the token-specific rotations. The issue is the pooled routing summary, not post-RoPE K itself: rotation and arithmetic averaging do not commute, and no one-center construction can preserve every token relation.

Centered subgists. For a chunk spanning $[p_{\text{start}}, p_{\text{end}}]$, the one-gist construction averages pre-RoPE keys and rotates the mean once at $p_c = (p_{\text{start}} + p_{\text{end}})/2$. For $G > 1$, we split the chunk into G balanced contiguous nonempty subspans, repeat this operation at each exact subspan center, and retain all subgist identities. Half-token centers are evaluated directly by RoPE rather than rounded; they occur in 30.4% of evaluated subspans. When a short chunk contains fewer than G tokens, G is capped at one nonempty subgist per token.

Chunk-ranking surrogate and protocol. For query i and candidate chunk c, we define the native-Q/K chunk-ranking surrogate

$$s_{ic}^{\text{native}} = \max_{j\in c}\left[\frac{1}{H}\sum_{h=1}^{H}\frac{q_{ih}^{\top}R(p_j)k_{icjh}^{\text{pre}}}{\sqrt{d_h}}\right]. \tag{9}$$

This scalar first averages token scores across heads and then maximizes across tokens. Actual multi-head attention normalizes scores separately within each head and combines values; therefore this is a chunk-ranking target, not an attention-output target or an approximation guarantee for the full attention operator. Every gist method uses the same mean-over-heads then maximum-over-gists reducer. We report the per-example Spearman correlation ρ_{QK} between its 15-chunk ranking and Eq. 9, plus top-1 and top-3 agreement. Thus "Q/K fidelity" below always means rank fidelity to this specific surrogate. Position bias is the per-example Pearson correlation between routing score and normalized chunk center; the table reports its absolute magnitude $|\rho_{\text{pos}}|$, while signed values remain in the artifacts.

The experiment reuses the frozen RoPE checkpoints, 16-way HotpotQA- and QASPER-derived partitions, established 80/20 split, tiny and small tiers, and seeds 1, 7, 21, 42, and 87. We compare post-RoPE mean, pre-RoPE mean, centered $G = 1, 2, 4, 8$, raw hidden cosine, and the supervised 32-D hidden-state projection introduced in the next section. Target-retrieval metrics use the programmatic evidence identity. Reported means first aggregate the four dataset–tier cells within each seed; 95% intervals use the five resulting seed means. The attention payload is always native token-level post-RoPE K and native V.

For clarity, the reported area is not ROC-AUC. The requested-fraction grid is

$$\mathcal{F} = \{.01, .02, .05, .10, .15, .20, .25, .30, .50, 1.0\}, \tag{10}$$

and, for the 15-candidate benchmark, $k(f) = \max(1, \lceil 15f \rceil)$. At each requested fraction f, $R(f)$ is mean held-out target-identity recall after selecting the top $k(f)$ candidates. We trapezoidally integrate the points $(0, 0)$ and $(f, R(f))$ for $f \leq .30$, then divide by .30. We call this the normalized target-identity recall/request-fraction area, $A^{\text{target}}_{0\text{–}30}$. Because ceiling is applied to 15 candidates, several requested fractions share the same physical cutoff; in particular, nominal 5% and 10% select 1/15 and 2/15, or 6.7% and 13.3%.

Table 4: Native-Q/K surrogate geometry and target-identity retrieval. ρ_{QK} includes a 95% confidence interval over five seed-level means. R10/R20 denote target-identity recall at nominal 10%/20% requested candidate fractions; $A^{\text{target}}_{0\text{–}30}$ is the normalized recall/request-fraction area defined in text, and f_{80}/f_{90} are actual selected fractions needed for .80/.90 recall.

Method	ρ_{QK}	$\lvert\rho_{\text{pos}}\rvert$	R10	R20	A^{target}	f_{80}	f_{90}
Post-RoPE mean	$.802 \pm .049$	.590	.631	.688	.653	.350	.523
Pre-RoPE mean	$.408 \pm .071$	.239	.527	.596	.548	.600	.683
Centered $G = 1$	$.794 \pm .052$	.589	.631	.692	.651	.373	.517
Centered $G = 2$	$.815 \pm .050$	.588	.627	.662	.639	.393	.537
Centered $G = 4$	$.847 \pm .040$	.597	.638	.692	.653	.373	.493
Centered $G = 8$	$.963 \pm .011$	.573	.627	.681	.652	.370	.483
Hidden cosine	$-.078 \pm .147$	.237	.100	.154	.131	.917	.963
Learned 32-D	$.178 \pm .058$	.338	.996	1.000	.971	.070	.073

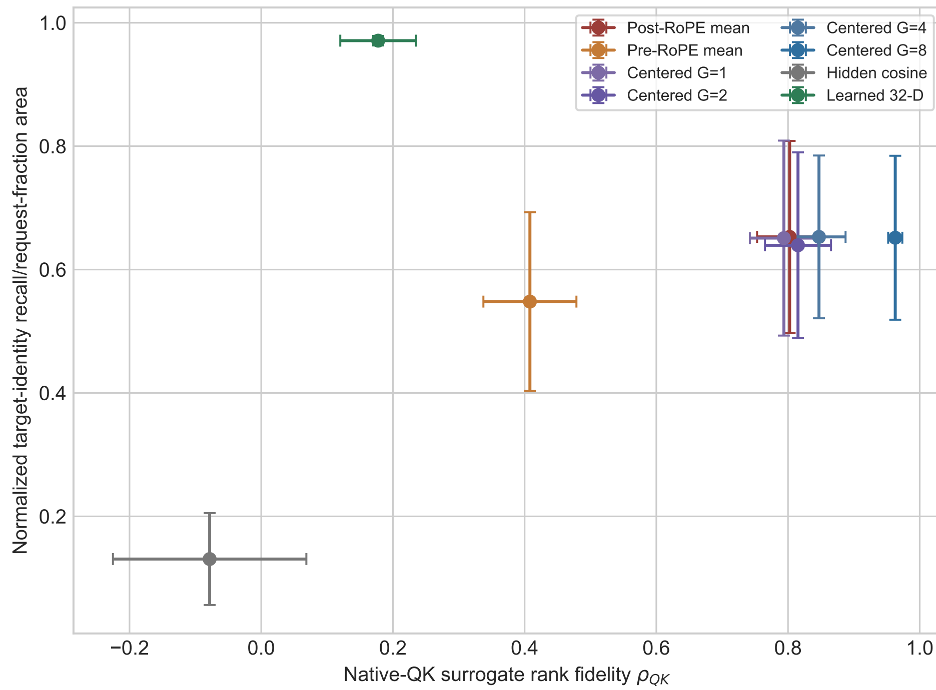

Figure 7: **Increasing centered-RoPE resolution more closely reproduces the defined native-Q/K chunk-ranking surrogate without improving programmatic target-identity retrieval; the supervised projection shows the crossed pattern.** Error bars are 95% confidence intervals over five seed-level means. The target-identity axis is the normalized recall/request-fraction area, not ROC-AUC.

The G sweep supplies the cleanest controlled contrast. From $G = 1$ to $G = 8$, mean ρ_{QK} rises by .169, and it rises in all 20 dataset–tier–seed conditions. Top-1 agreement rises from .723 to .946 and top-3 overlap from .715 to .923. Yet normalized target-identity recall/request-fraction area moves only from .6510 to .6516: 10 conditions improve, 8 decline, and 2 tie. R10 changes from .631 to .627 and R20 from .692 to .681. Diagnostic R@3/R@8/MRR are .692/.931/.653 at $G = 1$ and .681/.923/.662 at $G = 8$. Better approximation of the defined chunk-ranking surrogate therefore does not imply better target-identity ranking in this probe.

The learned metric provides the crossed comparison. It has only .178 surrogate rank fidelity, yet reaches .971 normalized target-identity area, .996 R10, and 1.000 R20. It is not an approximation of the native-Q/K chunk-ranking surrogate; it is a supervised target-identity geometry in this controlled construction. Centered $G = 8$ shows the reverse: .963 surrogate rank fidelity with only .652 target-identity area. These results experimentally separate the two measured ranking objectives in the paper's framework:

$$\boxed{\text{better native-Q/K surrogate ranking} \not\Rightarrow \text{better target-identity retrieval}} \tag{11}$$

$$\boxed{\text{high target-identity retrieval} \not\Rightarrow \text{native-Q/K surrogate similarity}} \tag{12}$$

For every method, fixed selected identities yield zero maximum absolute difference in selected native K, selected native V, and attention output. Pooling changes only the compact search representation. The result does not make token-level post-RoPE K wrong: native post-RoPE K is the

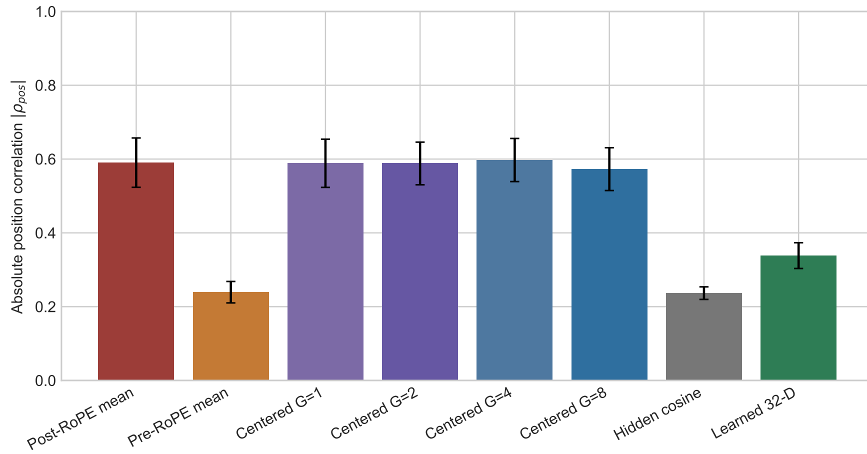

Figure 8: Native RoPE attention is position-sensitive, so increasing Q/K fidelity need not remove position correlation. Centered $G = 1$–8 retains substantial negative signed correlations ($-.569$ to $-.595$), whereas the learned target-identity metric reduces but does not eliminate the association. Error bars show 95% confidence intervals over five seeds.

correct attention payload. It shows why a mixed-phase mean can be a useful approximation of the defined Q/K ranking and still be the wrong objective for target-identity indexing. In particular, $G \uparrow \Rightarrow \rho_{QK} \uparrow$ does not imply $|\rho_{\text{position}}| \to 0$, because the target geometry is itself positional.

10 Can Routing Use a Different Geometry?

If increasingly faithful approximations of the native-Q/K chunk-ranking surrogate do not improve target-identity ranking, that controlled objective may require a different metric. We therefore test a positive alternative. Native RoPE remains responsible for attention, while a small metric with no explicit positional rotation at scoring time ranks chunks. Let h_q be the final query token's attention-input hidden state and g_c the mean attention-input hidden state of candidate chunk c. Two bias-free linear maps define

$$z_q = A_q h_q, \qquad z_c = A_c g_c, \qquad s_{\text{route}}(q, c) = \cos(z_q, z_c), \tag{13}$$

where $A_q, A_c \in \mathbb{R}^{32 \times d_{\text{model}}}$. The transformer, positional encoding, Q/K/V projections, and payload path remain frozen. Only A_q and A_c receive a within-example margin-ranking loss against the programmatic evidence identity.

> **A tiny supervised projection sharply improves fixed-anchor target-identity ranking without altering native positional K/V.**

Routing path

$h_q \to A_q \xrightarrow{\text{cosine}} A_c \leftarrow$ mean chunk h_{in}

$\downarrow$ chunk identities

Native attention path

selected identities $\to$ native post-RoPE K + native V $\to$ ordinary attention

Figure 9: Controlled target-identity selection and positional attention can use different geometries without changing the selected native K/V payload.

Protocol. We reuse the established 16-way HotpotQA- and QASPER-derived partitions, giving 15 candidate references per example, both model capacities, and five seeds. Each reference is encoded independently at its exact logical RoPE offset. We fit on the established 80% training split and evaluate the held-out complement. The four non-oracle comparisons are mean post-RoPE Q/K, mean pre-RoPE Q/K, untrained hidden-state cosine, and the learned hidden-state projection. A fifth router has identical initialization and optimization but within-example evidence labels are replaced by equally many non-evidence labels. The target is the benchmark's explicit answer-code evidence anchor; this isolates evidence-identity ranking but is not unrestricted question-conditioned QA retrieval.

A read-only audit of the frozen 64-example inputs exposes the available shortcuts. For both datasets, all 64 sources begin with `AnswerCode yes/no`, and the sole target is always `part-1`, the first of 15 candidates. The random split contributes 51 training examples and a 13-example held-out complement. HotpotQA has 51 distinct training source rows, 13 distinct held-out rows, and zero row overlap. QASPER has 48 distinct training papers and 13 distinct held-out papers, with one paper represented by different questions on both sides. Thus the QASPER split is not document-disjoint. The stored experiment has no position-only, random low-dimensional, anchor-only lexical, or randomized-location control. Attention-input hidden states receive no explicit rotation in Eq. 13, but may retain position- or construction-correlated information from the frozen contextual encoder. We therefore attribute the result only to supervised detection of the constructed target identity; which cue was learned remains unresolved.

Because 15 candidates are indexed, the requested R@5% and R@10% cutoffs realize one and two references, or 6.7% and 13.3%. Each example has one target identity, so any-evidence, evidence-identity, and all-evidence recall coincide in this probe.

Table 5: Held-out evidence-identity recall at the nominal 5% point (one of 15 candidates), mean over five seeds. Parentheses give the learned projection's population standard deviation. f_{80}/f_{90} are mean selected fractions required to reach .80/.90 recall.

Probe	Tier	Post-K	Pre-K	Hidden	Learned	Shuffled	f_{80}/f_{90}
HotpotQA	Tiny	.723	.446	.077	.908 (.031)	.015	.067/.080
HotpotQA	Small	.400	.446	.000	.985 (.031)	.000	.067/.067
QASPER	Tiny	.585	.369	.138	.938 (.090)	.000	.080/.080
QASPER	Small	.369	.369	.000	.969 (.038)	.000	.067/.067

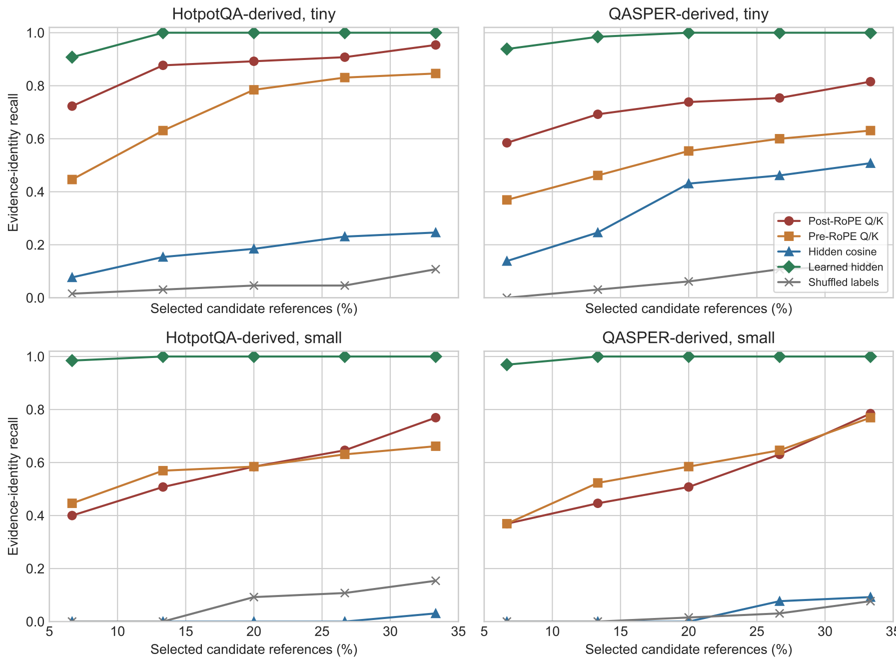

Figure 10: The learned 32-D hidden-state metric dominates raw hidden cosine and the shuffled-label control throughout the sparse range. Post-RoPE and pre-RoPE Q/K remain useful controls but do not define the best target-identity ranking geometry in this construction. Curves average five seeds.

At the two-reference point, learned recall is 1.000, 1.000, .985, and 1.000 in table order. Learned normalized target-identity area $A_{0\text{–}30}^{\text{target}}$ is .962–.980 and MRR is .954–.992. Its area exceeds raw hidden cosine and shuffled supervision in all 5/5 seeds of every dataset–tier cell. It also exceeds post-RoPE Q/K in all small-tier seeds and all QASPER-tiny seeds; HotpotQA-tiny is less uniform at 3/5 despite a positive mean difference. Thus the result is not that every learned initialization beats every K-space baseline. It is that evidence supervision exposes a target-identity ranking geometry absent from both raw hidden cosine and the matched label-noise control in this fixed-anchor construction.

Raw hidden-state cosine is an important negative result: one-reference recall is .000–.138, despite the learned maps receiving exactly those hidden states. The asymmetric projection exposes a more useful metric for this target identity. The shuffled-label projection collapses to .000–.015 recall, showing sensitivity to the assigned supervision rather than a generic benefit from low dimension or reparameterization. It does not exclude answer-code, candidate-location, or source-overlap shortcuts.

The router adds 4,096 parameters in the tiny tier and 8,192 in the small tier, or .31–.79% of the dataset-specific backbone. A 15-candidate float32 index falls from 3,840 to 1,920 bytes in the tiny tier and from 7,680 to 1,920 bytes in the small tier. The corresponding full native detail-K/V averages 52.9–55.3 KB and 105.9–110.6 KB. For a fixed selected identity list, post-RoPE K is identical, V is identical, and native attention output is identical: every comparison has zero maximum absolute error. The projection changes only the search representation, not the selected payload or its use by attention:

$$\boxed{\text{target-identity routing geometry} \neq \text{native attention payload geometry.}} \tag{14}$$

This closes the controlled comparison. Exact, local, fixed, clipped, and compressed RoPE placements did not yield a stable beneficial fixed distance. Increasing centered-pooling resolution did reproduce the defined native-Q/K chunk ranking, but did not improve target-identity ranking. Learned routing does not repair native attention geometry; it avoids asking the tested Q/K surrogate to serve as the target-identity objective. Larger contextual encoding improves the representation being summarized, learned projection improves controlled target ranking, and RoPE placement controls native positional attention. They are separate axes.

This contextualization result independently reinforces Paper 1's separation between encoding and routing granularity: fine addressability does not require equally fine independent encoding.

10.1 Does an Evidence Oracle Guarantee Better Use?

It does not. An evidence-only oracle is not universally best. Across dataset, tier, and position mechanism cells, the annotated evidence-only condition is worse than the router-selected set for 3.3%–78.3% of examples. In the continuous-head tiny-RoPE condition, moving from offset-plus-overlap routing to the constructed evidence oracle raises mean loss from 0.8763 to 1.0124 even though evidence recall becomes one. Annotated evidence identity is therefore not necessarily the functionally optimal context composition for a model. The result is a controlled warning, not a universal composition law.

11 Can Logical Prompts Exceed Native Operations?

The implicit `#__head` reference provides an intuitive long-prompt test. As the direct tail advances, displaced prompt tokens become one request-local source. Their logical positions continue rather than restarting at each rollover. The experiment uses $L_{\text{logical}} = L_{\text{position}} = 192$ and enforces $L_{\text{model-op}} = 32$.

Every measured encoding and attention call contains at most 24 tokens; native-limit violations are zero in all six tier–mechanism conditions. Source-relative offsets lower mean answer loss in 27/30 paired seed comparisons. Overlap lowers it further in 23/30. The effect size remains mechanism-dependent, and the small RoPE mean stays high despite the mechanical correction.

Table 6: Continuous-head mean answer loss over four examples and five seeds. All rows use 192 logical positions, a 32-token operation limit, and a maximum observed operation of 24.

Tier	Position	Reset routed	Offset routed	Offset + overlap
Tiny	Learned absolute	2.3837	.7304	.6474
Tiny	Sinusoidal	2.9691	.9388	.8021
Tiny	RoPE	2.5910	1.5547	.8763
Small	Learned absolute	3.8306	.6681	.0030
Small	Sinusoidal	5.0421	3.3860	.9110
Small	RoPE	6.7887	6.2249	3.0396

This demonstrates bounded positional and transport mechanics: the logical prompt exceeds one native operation, source-relative coordinates remain continuous, and no measured operation breaks its limit. It does not establish full functional equivalence to dense processing of all 192 tokens through every layer.

12 Pretrained Long-Context Position Controls

The controlled models establish the mechanism, but their short training range leaves open whether a reset remains consequential in pretrained long-context decoders. We therefore run pinned 4-bit Qwen3-8B, 14B, and 32B checkpoints on a 48 GiB M4 Pro. For each model, five unique examples from each of QASPER, HotpotQA, and 2Wiki are extended to 8K source tokens with deterministic same-dataset distractors. The full source is consumed either as an ordinary visible prefix, as segmented native K/V with its source positions preserved, or with the query incorrectly restarted at zero. The 8B boundary adds three 32K examples. The requested 64K rows are explicitly unsupported by the pinned 40,960-position models.

Table 7: **Query restart is far larger than unfused segmented numerical drift.** RMSE compares first-step logits with the ordinary visible prefix; agreement is exact greedy-sequence agreement.

Model	Context	n	source RMSE	restart RMSE	source agr.	restart agr.
Qwen3-8B	8K	15	0.112	2.572	0.667	0.000
Qwen3-8B	32K	3	0.128	2.530	0.333	0.000
Qwen3-14B	8K	15	0.100	4.786	0.933	0.000
Qwen3-32B	8K	15	0.121	3.671	0.667	0.000

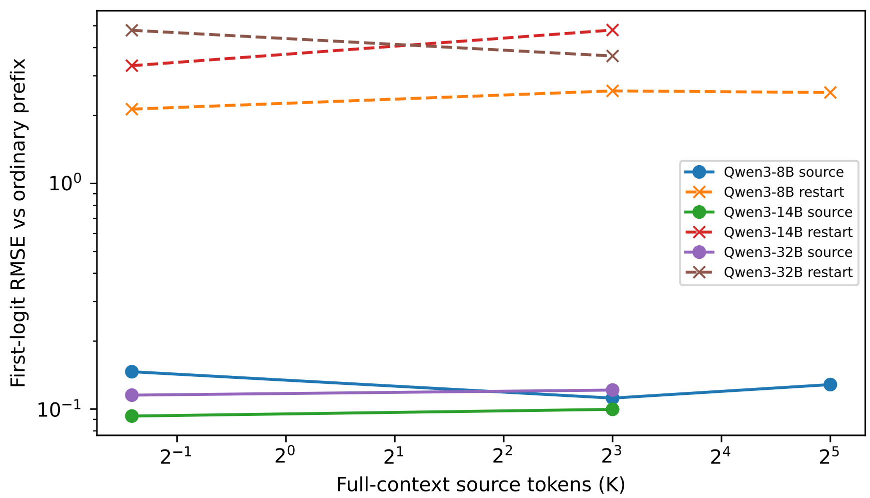

Figure 11: Preserving source positions keeps segmented-native logits close to the ordinary prefix. Restarting the query changes the geometry by orders of magnitude more at every measured scale.

At 8K, source-relative first-logit RMSE is 0.112, 0.100, and 0.121 for 8B, 14B, and 32B, versus 2.572, 4.786, and 3.671 under query restart. Source-relative sequence agreement is 0.667, 0.933, and 0.667; query restart is zero for all three. The three-case 8B 32K boundary has RMSE 0.128 versus 2.530 and agreement 0.333 versus zero. These are not claims that the segmented path is bit-exact. A companion concatenated-cache reference preserves 60/60 selected-text sequences across the Qwen size ladder and MoE checkpoint. That concatenated-cache result is shared experimental evidence with the standalone native-K/V manuscript, not an independent replication: both analyses use the byte-identical frozen artifact `shared/results/paper1_5_rope/mac_long_context/mlx_long_context_summary.json`. The residual source-relative error here comes from the unfused segmented one-softmax reduction repeated through many layers; the reset control is roughly 20–48 times larger.

For one contiguous source, global sequential and source-relative coordinates are structurally the same policy, so they are not counted as independent conditions. The negative control changes only the query base while preserving source tokens and native K/V. The experiment therefore strengthens the coordinate-correctness claim on pretrained models without testing arbitrary multi-resource serialization, retrieval-time rebinding, or semantic routing.

13 Related Work and Positioning

RoPE makes attention depend on relative displacement through position-dependent rotations [10]. Position Interpolation and LongRoPE extend dense-context operating ranges through rescaling and progressive extension [1, 4]. ALiBi and Transformer-XL provide other relative or distance-aware treatments [9, 2]. This paper is not another RoPE scaling recipe: it studies the ownership and rebinding of positions when K/V is encoded in chunks, stored externally, and retrieved sparsely.

Shortformer separates absolute positional information from reusable values [8]. Prompt Cache uses position-aware schemas for modular cache reuse, and CacheBlend selectively recomputes independently cached chunks that lack preceding context [5, 14]. MiniPIC similarly delays RoPE binding, while SemPIC learns a cache writer to address residual context dependence [7, 12]. These systems reinforce the distinction between position assignment and contextual hidden-state construction.

StreamingLLM preserves selected chronological decode states [11]. PRA instead routes among URI- or request-addressed chunks and must define both source identity and position semantics. Sparse attention and KV compression concern which state remains active or resident; the present question is whether a retained state still represents the coordinate and context its consumer assumes.

The controlled separation here motivates treating routing representation and native attention payload as distinct interfaces in pretrained-model retrofits; Paper 2 tests that transfer directly.

14 Limitations and Scope

The models are tiny and small controlled transformers, not pretrained production LLMs. The HotpotQA- and QASPER-derived tasks retain natural source text but replace open-ended answering with answer codes [13, 3]. WikiText-2 tests next-token behavior in bounded windows [6]. These choices isolate mechanisms but limit external validity.

The pretrained Qwen replication narrows that external-validity gap only for position transport. Its 8K model points contain 15 independent questions, the 32K point contains three, and 64K exceeds the pinned model limit. It uses one model family and quantization and does not establish quality-optimal rebinding or cross-family geometry.

Five paired seeds provide direction counts but weak inferential resolution: the smallest exact two-sided sign-flip result with five pairs is $p = 0.0625$. Formula-defined RoPE and sinusoidal coordinates can be evaluated outside the training range, but mathematical availability is not evidence of learned extrapolation. Learned absolute positions remain bounded by their table.

Overlap approximates missing left context through additional encoding and does not recreate a fully dense history at every layer. The evidence oracle is a programmatic source-identity control, not a guarantee of optimal context composition. Deferred-RoPE timing uses an unfused prototype path on one older GPU and should not be projected to production serving.

The fixed-distance and context-gate probes use scratch-trained tiny and small models with a 256-token training context. Their simplified answer-code loss reveals remote-placement failures, but small near-range deltas may not transfer to pretrained long-context models. Encoding-block size changes the causal context of later references, whereas materialized-unit size changes included neighboring content; neither is a pure scalar chunk-size intervention. The exact evidence-only context row is structurally invariant because evidence is encoded first.

The learned-router confirmation is deliberately narrow. Its positive identity is an explicit answer-code anchor at the beginning of every source and therefore always falls in the first candidate. Its train and held-out examples come from the same controlled dataset family; the QASPER split also shares one source paper across the two sides. It demonstrates that a tiny supervised metric can separate the mapped target from distractors without applying an explicit positional rotation in its routing score, not that it solves open-ended HotpotQA or QASPER retrieval. The shuffled-label control establishes sensitivity to the assigned supervision but does not exclude anchor, location, or document-overlap shortcuts. Position-only, random projection, anchor-only lexical, randomized-location, document-disjoint, hard-negative, and cross-domain controls remain untested. We do not retrain memory use or claim a downstream loss gain from replacing the selector.

The pooling-geometry experiment shares those evidence-identity and scale limitations. Its native-Q/K surrogate uses a maximum over token scores after averaging heads; actual multi-head attention instead normalizes per head and combines values, and other reducers can define a different ranking problem. Increasing G also increases compact-index entries and score work, so $G = 8$ is a fidelity control rather than a free serving recommendation. The study measures ranking and fixed-selection payload parity, not end-to-end answer quality after replacing the selector. Fixed-identity payload parity is a separate software invariant and does not establish attention-output approximation under changed selection. The dissociation does not make native Q/K geometry irrelevant; it concerns only the suitability of the stated mean-head/max-token surrogate for this target-identity search objective.

The paper deliberately excludes pretrained Hugging Face adaptation, learned memory-use training, recursive multi-hop retrieval, hidden-state multi-gist search, semantic-distance policies beyond the reported mechanism probes, and serving integration. Those extensions must not be read back into the present evidence.

15 Conclusion

The evidence supports the following conclusions. First, a source-relative logical start offset repairs reset fragmentation exactly at layer zero for learned absolute, sinusoidal, and RoPE models, and improves deeper representation in every paired test. Second, contextual fragmentation remains separate: overlap or grouped encoding can improve representation, but costs additional encoding and does not follow from correct coordinates.

Third, RoPE provides explicit algebraic Q/K rebinding, and matched pre-, post-, and deferred rotation preserve native attention geometry. That flexibility has a measured runtime cost, and no stable beneficial fixed retrieval distance emerged in the tested regime. Very remote placement can be destructive; the near-range response is non-monotonic rather than a universal distance rule.

Fourth, centered subgists increasingly approximate the defined mean-head/max-token native-Q/K chunk-ranking surrogate: rank fidelity rises from .794 to .963 in all 20 matched conditions from $G = 1$ to $G = 8$. Fifth, this geometric success leaves normalized target-identity recall/request-fraction area essentially unchanged at .6510–.6516. Sixth, a tiny asymmetric projection shows the crossed pattern, reaching .971 target-identity area with only .178 surrogate rank fidelity while selected native post-RoPE K, V, and attention output remain bit-exact for fixed identities. Because the answer-code anchor is always in the first candidate and shortcut controls are absent, this result establishes supervised target detection only in the reported construction. Seventh, annotated-target retrieval still does not guarantee downstream utility: annotated evidence and the functionally best context composition are not always identical.

The practical hierarchy is therefore: repair source coordinates, preserve contextually faithful representations, define native positional-attention geometry, choose and validate a task-specific routing geometry, retain native K/V for attention, and measure downstream use independently. **The metric appropriate for executing attention is not necessarily the metric appropriate for searching memory. Position, context, routing, and use are separate architectural problems.**

A Detailed Task Results

Table 8: WikiText-2 at an 8:1 logical-to-native ratio. Deltas are intervention minus baseline; negative loss is better. Every row contains five paired seeds.

Tier	Position	Δ final-K	Δ offset loss	Better	Δ overlap loss	Better
Tiny	Learned absolute	-.4281	+.0200	0/5	-.0173	5/5
Tiny	Sinusoidal	-.1897	+.0040	1/5	-.0007	4/5
Tiny	RoPE	-.6292	-.0000	2/5	-.0048	5/5
Small	Learned absolute	-.4044	+.0292	0/5	-.0140	5/5
Small	Sinusoidal	-.1810	+.0707	0/5	-.0102	5/5
Small	RoPE	-.6314	-.0000	2/5	-.0148	5/5

Table 9: Split-5 routed answer-token loss. Better counts paired offset-versus-reset seeds.

Dataset	Tier	Position	Reset	Offset	Offset + overlap	Better
HotpotQA	Tiny	Absolute	.559	.482	.482	4/5
HotpotQA	Tiny	Sinusoidal	.359	.304	.358	3/5
HotpotQA	Tiny	RoPE	.948	1.056	1.080	2/5
HotpotQA	Small	Absolute	.245	.043	.068	3/5
HotpotQA	Small	Sinusoidal	.061	.017	.012	4/5
HotpotQA	Small	RoPE	.129	.170	.196	1/5
QASPER	Tiny	Absolute	.629	.309	.301	4/5
QASPER	Tiny	Sinusoidal	.375	.244	.244	4/5
QASPER	Tiny	RoPE	.683	.869	.871	0/5
QASPER	Small	Absolute	.460	.299	.298	4/5
QASPER	Small	Sinusoidal	.403	.003	.007	5/5
QASPER	Small	RoPE	.169	.324	.184	1/5

B RoPE Storage Timing

Table 10: Small-tier CUDA timing in ms, averaged over five seeded runs of 500 warm iterations.

Storage	Selected K/V	Gather	Reconstruct	RoPE	Attention	Warm
Post-RoPE K	32	.0312	.0000	.0000	.1343	.1842
Pre-RoPE K + deferred	32	.0337	.0191	.4021	.1343	.6511
Post-RoPE K	160	.0327	.0000	.0000	.1342	.1791
Pre-RoPE K + deferred	160	.0329	.0145	.3838	.1342	.6487

C Implementation and Reproducibility

The implementation separates the native operation limit from position capacity. Each cached layer records token K/V, source identity, logical start/end, and whether K is pre- or post-RoPE. Contiguous token IDs are reconstructed as `logical_start + arange(length)`. Values are never rotated. Conversion copies the trained SelfAttention Q/K/V and output projections into PRA; with memory disabled, converted logits match the source model to test tolerance.

The final validation artifacts use commit `64f7cdf89b55`; generated cross-domain summaries use `052ae05`; the frozen distance/context run records source commit `03c9566ebbd7`. The original decomposition and storage matrix use `d36f67752701`. The learned-routing measurements use experiment implementation `d96c87b`; the pooling-geometry matrix records `527a50170802`. Each distance row records the checkpoint tokenizer fingerprint; this prevents a process-local tokenizer retraining from changing token IDs during evaluation. Raw CSV/JSON, compact publication CSVs, plots, expectations, and manifests are stored in the Paper 1.5 results directory. The reduction script regenerates manuscript artifacts without executing a model. The read-only split/location audit is `learned_routing/shortcut_audit.json`; the source-to-claim checklist and deferred experiment backlog are in `paper1_5_rope/CLAIM_LEDGER.md`. Tests cover large logical offsets, overlap and splitting, independent operation/position bounds, cache metadata, no-memory conversion parity, pre-/post-RoPE equivalence, deliberate rebinding, retrieval policies, rollover, and CPU/GPU parity. Learned-routing tests additionally cover projection shapes, backbone freezing, deterministic ranking, shuffled-label construction, fixed-selection K/V and output parity, and CUDA scores. Pooling tests cover balanced contiguous subspans, exact fractional centers, the $G = 1$ construction, short-chunk capping, deterministic correlations and top-k agreement, and routing-representation independence from native payload materialization.

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